Kutta-Joukowski Theorem: Derivation and the Generation of Lift

Ventura, César Carlos

Departamento de Física, UNS

Av. Alem 1253, (8000) Bahía Blanca, Argentina

E-mail: cesarventura.phys@gmail.com

Contents

1 Complex Potential Formulation **2**

1.1 Definition of the Complex Potential 2

1.2 Complex Velocity 2

2 Laurent Series and Far-Field Behaviour **3**

2.1 General Laurent Expansion 3

2.2 Far-Field Asymptotic Form 3

2.3 Identification of A_1 with the Circulation 3

3 Pressure Distribution and Force **4**

3.1 Bernoulli's Equation 4

3.2 Force on the Body 4

3.3 Evaluation of the Integral 5

4 The Kutta-Joukowski Theorem **7**

Abstract

This document presents a rigorous derivation of the Kutta-Joukowski theorem, a fundamental result in classical aerodynamics that relates the lift generated by an airfoil to the circulation around it. Starting from the complex potential formulation, we employ Laurent series expansions and the residue theorem to connect the circulation to the freestream velocity, Bernoulli's equation and Blasius' theorem then gives the force on the body, finalling arriving to the famous result $L' = \rho_\infty V_\infty \Gamma$. The derivation is presented step-by-step with detailed explanations, making it suitable for inclusion in a professional portfolio.

1 Complex Potential Formulation

1.1 Definition of the Complex Potential

For a two-dimensional, incompressible, irrotational flow, we define the complex potential

$$w(z) = w(x+iy) = \phi(x,y) + i\psi(x,y) \tag{1}$$

where ϕ is the *velocity potential* ($V = \nabla\phi$) and ψ the *stream function* ($V.\nabla\psi = 0$). The velocity components u and v are given by

$$u = \frac{\partial\phi}{\partial x} = \frac{\partial\psi}{\partial y} \tag{2}$$

$$v = \frac{\partial\phi}{\partial y} = -\frac{\partial\psi}{\partial x} \tag{3}$$

These are precisely the Cauchy-Riemann equations, so $w(z)$ is analytic wherever the flow is irrotational and incompressible (except at isolated singularities such as vortex centres).

1.2 Complex Velocity

The complex velocity is defined as

$$w'(z) \equiv \frac{dw}{dz}. \tag{4}$$

Because $w(z)$ is analytic, the derivative is independent of direction. Taking the derivative with respect to x

$$\frac{dw}{dz} = w' = \frac{\partial w}{\partial x} = \frac{\partial\phi}{\partial x} + i\frac{\partial\psi}{\partial x} \tag{5}$$

Hence

$$\boxed{w'(z) = u - i.v} \tag{6}$$

2 Laurent Series and Far-Field Behaviour

2.1 General Laurent Expansion

Since we are interested in the flow outside the body (a simply connected region), $w(z)$ is analytic there except for possible singularities inside the body. We expand $w(z)$ in a Laurent series about a point z_0 inside the body:

$$w(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n} \tag{7}$$

The coefficients are given by the standard integral formulas.

$$a_n = \frac{1}{2\pi i}. \oint \frac{f(z)}{(z - z_0)^n} dz \tag{8}$$

$$b_n = \frac{1}{2\pi i}. \oint \frac{f(z)}{(z - z_0)^{-n+1}} dz \tag{9}$$

More explicitly, the series has the structure

$$w(z) = \frac{A'_0}{(z - z_0)^m} + \frac{A'_1}{(z - z_0)^{m-1}} + ... + \frac{A'_{m-1}}{(z - z_0)} + A'_m + A'_{m+1}(z - z_0) + A'_{m+2}(z - z_0)^2 + ... \tag{10}$$

The negative powers form the *principal part,* the rest is the *analytic part.*

2.2 Far-Field Asymptotic Form

Far from the body, the flow approaches the freestream $w'(z) = u - i.v = V_\infty - i(0)$. Thus

$$w'(z) \to V_\infty \quad \text{as } |z| \to \infty \tag{11}$$

Therefore, all positive powers in the analytic part of w' must vanish. Setting $z_0 = 0$ for convenience, the complex velocity has the asymptotic expansion

$$\boxed{w'(z) = V_\infty + \frac{A_1}{z} + \mathcal{O}\left(\frac{1}{z^2}\right)} \tag{12}$$

2.3 Identification of A_1 with the Circulation

Integrating $w'(z)$ along a closed contour C that encloses the body:

$$\oint_C w'(z) dz = \oint_C \left[V_\infty + \frac{A_1}{z} + \mathcal{O}\left(\frac{1}{z^2}\right) \right] dz \tag{13}$$

The first integral vanishes because of $\oint_C dz = 0$. And given Cauchy's integral formula $\oint_c \frac{f(z)}{(z-a)} dz = 2\pi i f(a)$ and $\oint_c \frac{f(z)}{(z-a)^{n+1}} dz = \frac{2\pi i}{n!} f^{(n)}(a)$ we can calculate the second and third terms

$$\oint_c \frac{A_1}{z} dz = 2\pi i A_1 \tag{14}$$

$$\oint_c \frac{A_2}{z^2} dz = A_2 \frac{2\pi i}{2!}.(0) = 0 \tag{15}$$

Thus

$$\oint_C w'(z) dz = 2\pi i A_1 \tag{16}$$

On the other hand, from the definition of complex velocity,

$$w'(z)dz = (u - i\,v)(dx + i\,dy) = (u\,dx + v\,dy) + i(u\,dy - v\,dx) \tag{17}$$

The real part is $\mathbf{V}.d\mathbf{s}$, whose integral around a closed contour is the negative circulation $-\Gamma$ (convention dependent). For an incompressible flow, the imaginary part is the net flux through the contour $\oint_C \mathbf{V}.\hat{n}ds$, which is zero. Thus $\oint_C w'(z)dz = -\Gamma$

Comparing with the residue result gives

$$A_1 = \frac{-\Gamma}{2\pi i} \tag{18}$$

Therefore

$$\boxed{w'(z) = V_\infty + \frac{i\Gamma}{2\pi z} + \mathcal{O}\left(\frac{1}{z^2}\right)} \tag{19}$$

3 Pressure Distribution and Force

3.1 Bernoulli's Equation

For steady, incompressible, inviscid flow, Bernoulli's equation states

$$p + \frac{\rho}{2}(u^2 + v^2) = p_\infty + \frac{\rho}{2} V_\infty^2 \tag{20}$$

Since $u^2 + v^2 = |w'|^2$, we have

$$\boxed{p = p_\infty + \frac{\rho}{2}\left(V_\infty^2 - |w'|^2\right)} \tag{21}$$

3.2 Force on the Body

The force per unit span acting on the body is given by the integral of the pressure over the surface:

$$\mathbf{F}' = \oint_{\text{body}} p\,(-\hat{n})\,ds \tag{22}$$

To evaluate this integral using complex analysis, we must first justify the representation of forces in the complex plane. The physical force vector $\mathbf{F}' = (X, Y)$ is an element of the real vector space $\mathbb{R}^2$, where X denotes the streamwise component (drag) and Y denotes the transverse component (lift). The complex plane $\mathbb{C}$ is isomorphic to $\mathbb{R}^2$, this isomorphism maps the standard basis vectors $\hat{\mathbf{i}} = (1,0) \to 1$ and $\hat{\mathbf{j}} = (0,1) \to i$. Therefore, the force vector is represented by the complex number:

$$F' = X + iY \tag{23}$$

Now we express the contour element in complex form. The tangent vector to the surface is $dz = dx + i\,dy$. The outward normal vector in physical space is $(dy, -dx)$, which corresponds to the complex number $dy - i\,dx$. So:

$$dy - i\,dx = -i(dx + i\,dy) = -i\,dz \tag{24}$$

Substituting this into the force integral gives:

$$F' = X + iY = -\oint_C p\,(\hat{n}\,ds) = -\oint_C p\,(-i\,dz) = i\oint_C p\,dz \tag{25}$$

Thus the real part of this integral yields the drag, while the imaginary part yields the lift. Substituting Bernoulli's equation $p = p_\infty + \frac{\rho}{2}(V_\infty^2 - |w'|^2)$ and noting that the constant terms integrate to zero ($\oint_C dz = 0$), we obtain Blasius' theorem for the complex force:

$$\boxed{F' = -\frac{i\rho}{2}\oint_C |w'|^2\,dz} \tag{26}$$

3.3 Evaluation of the Integral

We now evaluate the integral appearing in Blasius' theorem. At first glance, one might attempt to evaluate this integral directly using the residue theorem, however, this is not possible. The integrand $|w'|^2 = w'\overline{w'}$ depends on $\overline{w'}$, the complex conjugate of the complex velocity. Since $\overline{w'}$ is a function of $\bar{z}$ rather than z, it does not satisfy the Cauchy-Riemann equations. Therefore, $|w'|^2$ is *not analytic* in the complex variable z, and the residue theorem cannot be applied.

Blasius' theorem provides an elegant resolution to this difficulty. It exploits the fact that the airfoil surface is a streamline, which allows us to relate the non-analytic integral of $|w'|^2$ to an analytic integral of $(w')^2$.

Since the contour C lies on the airfoil surface, it is a streamline of the flow. Therefore, $\psi =$ constant and $d\psi = 0$ along C. Thus:

$$w' = \frac{dw}{dz} = \frac{d\phi}{dz} + i\frac{d\psi}{dz} = \frac{d\phi}{dz} \tag{27}$$

which is purely real. Consequently, on the contour:

$$\overline{w'\,dz} = w'\,dz \tag{28}$$

Multiplying both sides by w' gives:

$$|w'|^2\,\overline{dz} = (w')^2\,dz \tag{29}$$

Integrating around C and taking the complex conjugate yields the following identity:

$$\boxed{\oint_C |w'|^2\,dz = \overline{\oint_C (w')^2\,dz}} \tag{30}$$

The crucial advantage of this transformation is that it can be evaluated using the residue theorem. squaring the asymptotic expansion of the complex velocity we obtain:

$$(w')^2 = V_\infty^2 + \frac{i\,V_\infty\Gamma}{\pi z} + \mathcal{O}\left(\frac{1}{z^2}\right) \tag{31}$$

Now we evaluate $\oint_C (w')^2\,dz$. Since $(w')^2$ is analytic in the region between the airfoil surface and infinity, we can deform the contour C to a large circunference of radius $R \to \infty$ without changing the value of the integral:

$$\oint_C (w')^2\,dz = \oint_{|z|=R} \left[V_\infty^2 + \frac{i\,V_\infty\Gamma}{\pi z} + \mathcal{O}\left(\frac{1}{z^2}\right)\right] dz \tag{32}$$

We evaluate each term using. Just as before, the first and third term vanishes. The second term gives: $\oint_{|z|=R} \frac{i\,V_\infty\Gamma}{\pi z}\,dz = \frac{i\,V_\infty\Gamma}{\pi} \cdot 2\pi i = -2\,V_\infty\Gamma$.

Therefore:

$$\oint_C (w')^2\,dz = -2\,V_\infty\Gamma \tag{33}$$

Since this result is real, substituting this result into Blasius' theorem:

$$F' = -\frac{i\rho}{2}\oint_C |w'|^2\,dz = -\frac{i\rho}{2}(-2\,V_\infty\Gamma) = i\,\rho\,V_\infty\Gamma \tag{34}$$

Thus:

$$X + iY = i\,\rho\,V_\infty\Gamma \tag{35}$$

Equating real and imaginary parts gives the Kutta-Joukowski theorem:

$$\boxed{X = D' = 0} \tag{36}$$

$$\boxed{Y = L' = \rho\,V_\infty\Gamma} \tag{37}$$

4 The Kutta-Joukowski Theorem

We have obtained the fundamental result:

$$\boxed{L' = \rho_\infty V_\infty \Gamma} \qquad \boxed{D' = 0}$$

where

- L' is the lift per unit span (positive upward).
- D' is the drag per unit span.
- ρ_∞ is the freestream density.
- V_∞ is the freestream speed.
- Γ is the circulation.

This is the **Kutta-Joukowski theorem**.